

## 4 Making it a science

### Aspirations and apprehensions of transitions research

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#### 1 Introduction

Making transitions research a science should be, in my opinion, the highest priority on the agenda of the field. Higher than, for instance, striving for policy relevance and influence. Indeed, one could argue (as I do) that it is questionable to strive for such things while transitions research is not yet scientific. Note that I am not only saying that the body of knowledge to have come from transitions research is not yet mature enough to be called science. I am saying that transitions research at the time of writing is itself, by and large, not scientific or even unscientific.

I expect there to be many transitions researchers who wholeheartedly disagree with the paragraph above. Moreover, I expect them to divide neatly into two groups who disagree in two complementary ways. One group would argue that transitions research does not need to become a science because transitions are not necessarily a scientific subject and therefore cannot be treated scientifically. Another group would argue that transitions research is scientific research already, that it is social science and while this entails a different suite of methods and approaches it is science nonetheless.

To the first group, transitions must be something like art – and I do not say this to ridicule – and transitions research a bit like art critique. Their thinking seems to be along the lines of: while art can be studied scientifically, the appreciation and pursuit of art are at best marginally aided by this. If you identify with this group, I disagree with you and most likely you with me, but our disagreement is one of general principle and I do not expect it to be amenable to argument.

To the second group, transitions (and likely various other subjects) belong to a special class of phenomena: the human phenomena. These researchers would see a range of reasons why the standards and practices of science (which they would call ‘hard science’) do not apply in their field. Amongst their reasons are things like the unpredictability of human action, normativity, (inter)subjectivity and so on. If you identify with this group, I disagree with you and most likely you with me, but our disagreement is more subtle and I will try to make an argument for my case in Section 2.

### 1.1 *The importance of transitions*

That transitions are important is perhaps the one notion that unites transitions researchers in their endeavours. Transitions, to me, are important in several ways, but the unifying importance is *practical*. I usually summarise this importance with reference to *persistent problems* and *pervasive change*.

Persistent problems are, in a way, the negative side effects of the systems we rely upon to meet our human needs on societal scales. Rotmans (2005) sees persistent problems as symptoms of an unsustainable society. Schuitmaker (2012) points out that persistent problems are not so much malfunctions of the systems but undesired consequences of the way they work. From this perspective, transitions are ways to overcome persistent problems. If the systems are intrinsically problematic, then the systems need to structurally change.

Pervasive change seems to come closer to the way transitions were described in the pioneering literature. The classic *More Evolution than Revolution* (Rotmans et al., 2001), for example, refers to transitions as the confluence of developments in several domains, leaving open whether these developments are purposefully pursued or ‘emergent’. Pervasive change is thus the transitions that happen to us whether we want it or not. From this perspective, transitions are something to deal with. Positively speaking they present opportunities. Negatively speaking they need to be navigated to mitigate potential harm.

Whether we ‘merely’ want to navigate a transition or purposefully pursue one, it seems obvious that we need to *understand* them, and I think this is the unifying importance that is a major motivation of most, if not all, transitions researchers.

Even if one is not interested in the possibility to navigate or pursue transitions, or if one is deeply skeptical about the possibilities to do so, then still there is the noble quest to understand the movements of the societies we are part of. The changing systems we have created to sustain us and of which we are a part, determine a large extent the modern human condition. Transitions are therefore eminently worthy of study as a contribution to human culture at large. Even if one does not appreciate these lofty connotations, then there is still curiosity. One wants to know because one does not yet. Because of the pleasure of finding things out, as Feynman (1999) would have it. Perhaps because of the beauty of what we will find. Perhaps because we cannot help but want to know. Transitions are a Grand Intellectual Challenge.

### 1.2 *The desirability of a transitions science*

Curiosity and the Grand Intellectual Challenge are enough to make me desire a transitions science, regardless of its utility. To paraphrase a quote generally attributed to<sup>1</sup>, again, Feynman: science is like sex, it has practical applications

but that's not why we do it. But for those of you who also want transitions research to matter to actual transitions (likely most researchers in the field, as per my assessment in Section 1.1), I ask: why is there so precious little scientific research on transitions?

Leaving aside, for just this moment, whether the dominant mode of transitions research produces knowledge that *can* matter, one really wonders why a field so concerned with real-world issues like sustainability does not embrace more scientific modes of operating. Is it a perverse irony of this field? That its 'regime' of narrative case study based research is so strong it discourages budding transition scientists? Or worse? It is not difficult to recognise as downright hostility the not-even-thinly-veiled patronising tone of regime advocates like McDowall and Geels (2017). While that article was a response to Holtz et al.'s (2015) position paper on modelling, my estimate is that their response to anything even more explicitly in favour of a 'hard' science approach would invoke even stronger corrective responses and scorn – especially given their antipathy to anything reeking of 'positivism'.

I can think of no other field that matters, in the sense that transitions research could, putting its trust so narrowly in an approach not known for yielding reliable, accurate knowledge. Medicine and psychology may share the ubiquity of case research with transitions, but these fields also employ laboratory experiments, modelling and randomised trials while the case research is routinely meta-analysed and a single case report is likely to be viewed as mere anecdotal evidence. I mention these two fields also because they share many of the complexities, ethical implications and empirical difficulties of transitions research<sup>2</sup>. There is surely a role for case-based research – narrative or otherwise – in a science of transitions, which I will discuss extensively in Section 3.6. The problem is that at this point, the transitions research play seems written as a monologue.

It is of no relevance whether or not the objects of study in those fields are more amenable to 'hard' scientific methods. The point is that if we do not know anything scientifically about such issues, we know very little at all and whatever we opine or may think they mean is no basis for policy or otherwise intervention. Imagine public health services preparing for a pandemic solely on the basis of detailed historical accounts of previous outbreaks. Not that such accounts are wholly inappropriate for future strategy and policy – if only to learn from obvious mistakes – but what is needed, what the public rightfully expect, is that strategy and policy are based on a thorough, tested, mechanistic understanding of how disease is contracted, propagated and treated. It is ethically objectionable to even suggest important decisions be informed by transitions research if not scientific.

### 1.3 Science as aspiration

I have pointed out several times already that most transitions research is at the moment not scientific. Precisely what attitudes and practices are problematic

in this sense, I will elaborate extensively in Section 3. As a matter of course, since this seems to be a strictly negative assessment, I will outline concrete steps that can be taken right now by transitions researchers to make it a science. This I will do in the concluding part of this chapter (Section 5), based on my — philosophically uncontroversial — interpretation of the intimate relation between theorising, models and empiricism (Section 4), taking practical cues from other fields that face similar challenges. Before dismantling the arguments against the possibility of a transitions science (Section 2), I will now sketch what I — and I believe most scientists — consider the nature and features of scientific research.

Despite the household notion of the ‘scientific method’, science is *not* a matter of method. Advanced sciences have advanced methods of course, but this is a consequence of their progress, not the reason for it. As their theories become more accurate, they need more precision<sup>3</sup> to know whether observations match those theories or not. If science is not a matter of method, then what *is* it a matter of? It is a matter of honesty. An intellectual honesty going far beyond not wanting to lie. It is the acceptance of the possibility of being wrong, moreover it is actively setting oneself up to be proved wrong constantly. Feynman (1974; 1985; 1999) calls it ‘a kind of leaning over backwards’ and urges publications of ‘negative’ results as well as ‘positive’. Dennett (2013, p. 27) says<sup>4</sup>: ‘In science you make your mistakes in public’.

Proving a hypothesis wrong also progresses knowledge, perhaps even more than finding confirming evidence does. So, rather than cushioning conjectures in vague terms so no-one is ever quite sure what is stated or denied, we should acknowledge that we are conjecturing and phrase ourselves refutably. This prescription is usually attributed to (Popper, 1957) though it just summarises the scientific attitude. One of the very worst things is to be ‘not even wrong’, which Pauli reputedly liked to point out. Theoretical speculation, conjecture and conceptual design are of central importance in scientific research, the above is certainly not meant to decry speculation as such. In fact, one sees its importance acknowledged in the explicit division of labour in some advanced sciences, where theoretical research is represented by specialised branches of otherwise empirical fields. The art of conjecture is to say something that is not trivial, that is, not just a label for what is empirically known, and that could be false.

It is important to appreciate the role of method correctly, because wellintended scientific aspirations can easily lead to naïve imitation of methods common in the natural sciences. Minsky (1998) referred to this tendency as ‘physics envy’. This may lead to what Feynman (op. cit.) has described as ‘cargo-cult science’<sup>5</sup>, in which methods of the ‘hard’ sciences, like quantification and statistics are applied while missing the point of being scientific, and therefore the rationale of those methods, completely. It is not that they apply those methods badly, nor is it that they do not apply at all, that may or may not be the case. The point is that using such methods does not *make* the research scientific.

Science seeks understanding of classes of phenomena and this understanding is considered to be brought about by *explanation*. Hence philosophy of science has since long had a strong focus on the nature of explanation. Theories, from

this perspective, can be thought of as bodies of meta explanations, generic formulae that, when filled-out with the specifics of the situation at hand, explain the phenomenon of concern. Hempel's Covering-Law Model (see e.g. Hempel and Oppenheim, 1948) is perhaps the best known example, but later models based on causal mechanisms and functional explanation (see e.g. Salmon, 1990) are in this sense quite similar.

This perspective makes generalisation a key part of any scientific explanation. To explain something first requires one to identify it as an instance of a class of phenomena. One then invokes the law or mechanism about that class of phenomena with the parameters pertinent to the specific phenomenon under consideration. If the observed now logically follows, it is explained.

In other words, *no explanation without generalisation*. This seems in stark contrast with historical explanations where there are only particulars and contingencies. Such explanations usually rely on the course of events to provide the explanation. Every next event is expected given the previous ones in the sequence. Implicitly, every step involves an elliptical invocation of a law or mechanism considered so obvious it can go without saying (e.g. Roberts, 1996, for an explication of this view of historical explanation). If done well, this may satisfy as an explanation of a particular event or case (but see Section 3.4 for common pitfalls), it does not typically contribute to understanding the class of phenomena the sequence of events was a particular instance of.

This perspective also makes the capacity of *prediction* a central desideratum of any scientific explanation. Amongst natural scientists, this would not cause much of a stir, but I expect researchers of social phenomena may object that this is too much to ask – or that it is not of central importance. Is it really? 'Any rationally acceptable answer to the question "why did X occur?" must offer information which shows that X was to be *expected* – if not definitely ... then at least with reasonable probability', said Hempel (quoted in Rosenberg, 1993, my emphasis). I feel this is a fair, if not modest, demand. This capacity of prediction is also precisely what makes theories testable. Moreover, it is this predictive aspect that makes science *matter* in the way transitions research wants to matter.

That science is empirical is one of its defining characteristics. Yet there are common misunderstandings about the role of empiricism in science. It does patently not mean, for instance, that scientific hypotheses or theory are always obtained inductively. Spotting a pattern inductively is definitely a way to come to a hypothesis, but there are infinitely many hypotheses for any one pattern observed so to say induction *leads* to a hypotheses seems to strong. The practice of analytic theorising is much more a form of induction-inspired speculation. Moreover, it is often the noticing of something *not* fitting a pattern that gets the hypothesising going.

How science is empirical and the *practical* relation of theory and experiment in sound science have rarely been more accurately summed up, likely never with less jargon, than by Feynman. In his lectures on the character of physical law (1965), he said: 'In general, we look for a new law by the following process. First, we guess it. [LAUGHTER] Then, we compute – well, don't laugh, that's

really true. Then we compute the consequences of the guess [...] we see what it would imply. And then we [...] compare to experiment or experience. [...] If it disagrees with experiment, it's wrong. And that simple statement is the key to science. [...] That's all there is to it'.

The empirical test provides science with an ultimate arbiter. Clearly, the guessing is not arbitrary, it is educated and may take cues from established theory or other considerations, like elegance. Clearly also, where Feynman says 'compute', one can safely substitute 'infer' – it is the logical implications of our hypotheses that matter, they endow hypotheses with the predictive powers that make them testable. Though Feynman was a physicist and the quoted lectures were on the character of physical law, his summary of the practice of good science is noticeably devoid of any reference to a particular empirical domain. In fact, he literally speaks '[i]n general' about 'science' and I do not see any reason to restrict his lessons to physics or even the natural sciences (see also Section 2). Feynman's lesson also does not refer to any specific method. Not for the experimenting, not for the guessing, nor for the way one should go about comparing guesses to experiments. It is not a matter of method.

## 2 The case against a transitions science

I mentioned in the introduction that there are numerous transitions researchers who believe that the standards and practices of what they would refer to as 'hard' science do not apply to their field – the people I referred to as 'the second group'. Defending this belief they would bring to bear various technical arguments against the possibility of a true transitions science. I do not disagree with all the issues they raise, but I think their conclusion does not follow. I will try to distill the common kernels of their arguments and show how I think they are *non sequitur*. Apart from these technical arguments there is a common non-argument based on misguided ideas about philosophy. I will discuss this also.

Whatever may be said in favour or against these arguments, I feel that there are 'deeper' reasons for people to oppose a scientific approach to transitions. These reasons, I think are based on a misguided perspective on the social sciences vis-à-vis the natural sciences – namely that there would be a separation between the two and, moreover, that this supposed separation would warrant different standards and methods. I will point out what I think are the origins of this perspective and urge to move beyond it.

### 2.1 Technical arguments

The supposed reasons against a 'hard' scientific treatment of transitions can be grouped into the following three categories:

*Too complex* – According to this argument, societies, or at any rate the dynamics of transitions, are just too much for the crude methods of science. The generalisations that the scientific approach is after, require simplifications that make a caricature out of the rich fabric of historically

unfolding transitions. This latter point is no doubt true, but out of many (empirically tried and tested) caricatures we build scientific theories: robust generalisations across a class of phenomena. The argument from complexity did not stop biology and medicine from charting and understanding organisms and human bodies, which are stupefyingly complex systems composed of tens of *trillions* of cells, which are themselves very complex systems. Organisms have functional structures and dynamics on and across many scales, covering about 20 orders of magnitude (see e.g. West et al., 2002). Certainly, this presents a massive challenge, but the continuing progress of chemistry, biology, genetics and medicine shows that complexity of arguably larger<sup>6</sup> proportions can be understood with scientific methods. Such progress should inspire transitions researchers.

*No lessons from the past* – This argument roughly says that whatever hope one has of generalisations with predictive power, such generalisations would be intimately tied up with the historically contingent structures of the societies they would be found in. These structures themselves are prone to change (transitions!) and then those generalisations no longer apply. In other words, whatever we learn about transitions in the 20<sup>th</sup> and 21<sup>st</sup> Centuries will be void when a new socio-politico-economic era dawns, perhaps heralded in by the very transitions we pursue. Karl Popper (1957) has put forward such an argument and Rosenberg (1993) explores it to identify limits to prediction in social science. The argument seems to me not flawed and Martínez Arranz (2017) raised it in the context of transitions.

The in-principle impossibility Popper raises, which Rosenberg discusses, I find not very intimidating – at least not for the prospect of a transitions science. The notion that we can never predict e.g. future technological advances (because it would imply the possibility to extrapolate future knowledge from current knowledge, which implies we *already* have the knowledge required to attain it) is amusingly paradoxical. However, such a science of technological progress would meet limits of prediction much earlier. Even in Newtonian physics where such *inprinciple* impossibilities are absent (remember Laplace's demon), practicalities like non-linearity, measurement uncertainties and many-body problems lard the ointment with flies. The ambitions of a transitions science, though storming the heavens, would be much more modest than that.

*Impossibility of mind-reading* – This last category of objections against a scientific approach to transitions, argues in the following fashion: Transitions are the product of human actions, and therefore we have to deal with the intricacies of the human mind, like issues of agency, (inter) subjectivity and meaning. Something the 'hard' sciences do not have to, or refuse to deal with. This argument is flawed in two separate ways. Firstly, issues of agency, subjectivity and meaning *are* amenable to scientific inquiry. The interdiscipline of cognitive science has made serious headway in these areas and again this should inspire transitions researchers. The second way in which the argument is flawed is this: the

(supposed) fact that human action is contingent on subjective perceptions and meanings attributed, does patently *not* mean that they do not display regularities that can be captured in ‘hard’ science-like covering laws – which can even be in terms of subjective perceptions or meanings if appropriate. Behavioural science is certainly finding that human actions are quite predictable. In fact, everyone routinely ‘reads mind’ all the time in social situations and deviations from our ‘predictions’ give rise to humor or affront. Moreover, these issues may matter much less than we are inclined to think – I will return to this shortly.

## **2.2 *Philosophy non-argument***

Some object that a ‘hard’ scientific approach in the vein of what I am suggesting is based on a different philosophy of science, one that may not be appropriate to social sciences. This is of course just another form of the separatist argument of ‘the second group’. In this specific form, however, it also reveals a thorough misunderstanding of the role of philosophy of science. Many social scientists seem to be under the impression that one has to *choose* a particular philosophy of science, like if it were methodology and one simply has to find the one best suited to the topic. This is clearly mistaken.

Philosophy of science does not aim to be a foundation of science, it aims to clarify what science is. One of the ironies of philosophy of science is that it is acutely aware of the fact that science has been getting by without philosophy very well thank you. Moreover, the successes of empirical science ‘sans philosophy’ inspired early analytic philosophers and philosophers of science, like Russell and the Vienna Circle to remodel philosophy in the image of science – *not* the other way around.

This misunderstanding leads to the absurd situation of graduate students having to write perfunctory thesis chapters where they declare their allegiance to e.g. pragmatism as a supposed fundament and justification for their mixed methods approach. Clearly, whoever uses statistics or tries to measure something operates in a positivist tradition, while qualitative researchers are shepherded towards interpretivism. I think this is a real and entrenched problem in academic research training, perpetuated by popular ‘research design’ textbooks (e.g. Creswell (2018) – five editions and over 100,000 citations on Google Scholar) that artificially distinguish philosophical ‘worldviews’ and identify them with certain methods.

Another way in which philosophy is commonly misconstrued in transitions research and other social sciences is through terminology. The terms ‘ontology’ and ‘epistemology’ are routinely used to mean pretty much ‘concepts’ and ‘methodology’. Ontology is a branch of metaphysics<sup>7</sup> interested in what the world ultimately consists of, e.g. matter or mind, and what that may mean for the existential status of such things as universals. Epistemology is theory of knowledge. What constitutes knowledge? How can we be sure that something is actually knowledge?

To refer to one’s conceptual framework as an ‘ontology’ suggests a metaphysical commitment to those concepts being ‘really out there’. The scientific



modus operandi is to not care what the metaphysical status of a concept is. What is the metaphysical status of centres of gravity? Do species really exist<sup>8</sup>? What shall it matter? To refer to one's methodological approach as an epistemology is also confused, albeit in a slightly different manner. It is mistaking the means to acquire something (here, knowledge) with the nature of the something. Like saying, work is the essence of money. Discussing ways of obtaining money is simply distinct from what money is and is not – unless one wants to get *really* philosophical about it. This is not to say there are no epistemological concerns about transitions worthy of philosophical inquiry, just that when transitions researchers use the term, typically they mean something else.

### 2.3 Deeper reasons

I think that underneath the technical objections raised against the possibility of a transitions science lurks something deeper, something rooted in fear, a twofold fear.

The first is psychological, it is fear of inadequacy. This would apply to those who feel a scientific approach would actually be superior, but lack the requisite knowledge and skills – or the motivation to obtain them. The resulting cognitive dissonance is alleviated by adjusting attitudes, towards supposing 'hard' science approaches to not apply to their topic. Perhaps the prevalence of this fear is not that great. At any rate, it is unnecessary, see my comments about 'no matter of method'.

The second is philosophical, it is the pre-Copernican fear of us humans losing our special place in the universe. Especially the third genre of technical objections, with its deference to mind as the great separator, is a child of this fear. I believe the idea of a sharp separation between human phenomena and the rest of the world, and its concomitant necessity of special modes of inquiry, to be an echo of antiquated ideas with a shared core of human self-importance: The sharp separation between earthly and celestial mechanics – an idea no longer tenable after Newton. Humans as distinct, and placed above, animals and the rest of nature – ridiculous after Darwin.

'Life and all its glories are thus united under a single perspective, but some people find this vision hateful, barren, odious. They want to cry out against it, and above all, they want to be magnificent exceptions to it', said Dennett (1995, p. 144). But we *are* part of the animal kingdom, part of the great Tree of Life and part of the fascinating physical universe in which it sprouted. We should take this at face value in our quest to understand transitions.

Even if we come to learn that certain species of birds have a rich inner life and complex cultural interactions based on the meanings they attribute to aspects of their world, I doubt we would then conclude that scientific methods and explanations no longer apply. Nor would we conclude they then transcend biology, that their behaviour and social structures are then somehow of a totally different order. Fields like sociobiology (Wilson, 1975) and subsequently evolutionary psychology (Symons, 1979; Tooby and Cosmides, 1992;

Barkow et al., 1992) have taken Dennett's single perspective at face value, and indeed attracted the expected controversy and scorn – so much for the intrinsic unpredictability of human behaviour.

'The social sciences need to take seriously their status as divisions of biology' and they therefore 'must proceed in accordance with the explanatory and evidential strictures of biology', says Rosenberg (2017, p. 341–342) in a most lucid recent piece of philosophy of (social) science. He makes abundantly clear how this is not a reductionist thesis in neither of the common meanings of the term 'reductionist'. It is the practical observation that the methods of the social sciences should be no different than those of biology, and moreover, that the 'findings, theories, and models of human behavior and of human groups [would be] no different in kind from those biology is familiar with'.

That this is not a singular voice advocating an isolated scientificist day-dream can be gleaned from recent developments. Fields that traditionally looked at human phenomena in isolation from the rest of nature or did consider other animals as part of their jurisdiction at all, begin to adopt the single perspective. An example of the former, is the research on the evolution of leadership looking at the human manifestations of the phenomenon on equal footing with those of other animals (e.g. King et al., 2009; Van Vugt, 2006). An example of the latter is how some archaeologists have broadened the scope of their discipline to study the material cultural history of non-humans (Haslam et al., 2009, 2016; Proffitt et al., 2016; Wong, 2017) —this should be especially inspiring to transitions researchers as it entails the disappearance of the artificial distinction between human and non-human technological innovation. These developments routinely find their way to major publications of the natural (*Nature*, *Current Biology*) and 'human-centred' sciences (*Personality and Social Psychology Review*, *Journal of Human Evolution*).

### **3 What is wrong with transitions research?**

The practices I will raise in this section are not necessarily problematic in principle. As I mentioned in the introduction, science is not a matter of method. That, however, does not mean that every method is as good as the next. Some are certainly worse than others. Some are so because they invite a way of thinking that almost precludes a scientific approach. Whenever this is the case, I will attempt to clarify how this happens.

In the following, I will discuss practices that I think are pervasive in the field to the point of being typical. I am certainly not suggesting that all researchers employ them, nor that each researcher employing them is necessarily being unscientific. The point is more subtle. Because of the prevalence of these practices in the field, people do not question their legitimacy from a scientific point of view or otherwise. A whole community cannot be wrong, right? Therefore, the dissenters are wrong. Their approach must not apply.

### 3.1 Fetish of the peculiar

Plainly put, much research on transitions tries to explain what is *special* about a particular transition or part of it, whereas the aim should be to find out what is *common* across transitions. This is not to say that an explanation of what seems to be an exception cannot be enlightening. Indeed, often scientific theories are illustrated by how they subsume supposed exceptions under a general principle.

This is, however, not the kind of explanation-of-the-peculiar common in transitions research. There, general principles have often not yet been found and thus explanations of apparent exceptions will necessarily be accounts of the particular contingencies involved. At best these will actually constitute explanations, but it would be a failure of any scientific theory of transitions if many such exceptions remain to be explained in this manner.

This lure of peculiarism is great as what stands out cries for explanation. It is amplified by the large reliance on case studies as the source of empirical material. The virtue of case studies is that they provide in-depth, in-detail accounts of phenomena but this virtue tempts to forego the search for generalisation. Even amongst the comparative case studies around, the tendency is to focus on what sets them apart.

The fetish of the peculiar is self reinforcing. The more the focus is on what is special, the less we search for general principles, the more we lack scientific explanations for apparent exceptions, the greater the reliance on peculiarist explanation.

### 3.2 Fuzzy concepts that move

There is nothing problematic about concepts being fuzzy or their interpretation shifting over time. Indeed, it is quite common for the central concepts of a science or discipline to be vaguely defined. Biology is not troubled much by the lack of a precise definition of 'life' or 'species' and physicists do not bat an eyelid about shifting back and forth between an interpretation of space as 'empty expanse' or 'stuff' that can bend and curve. So why worry about concept fuzziness and motion?

Sometimes it seems that more precise definitions are a sign of maturity of the field (e.g. the concrete definition of 'genes' in terms of DNA molecules), but this precision may just as well be a consequence of the progress as the cause. Moreover, I think that pedantically emulating the conceptual precision of the exact sciences will lead to a theoretical variant of the cargo-cult science I discussed in the introduction. The point is that conceptual precision has to be earned – empirically.

When we are still in the explanatory dark, our concepts in fact likely *need* to be a bit fuzzy, though obviously not as fuzzy as to mean anything and everything. The testing of hypotheses, however, necessitates quite precise operational definitions of concepts to unambiguously know whether something is an instance of the class of phenomena we are trying to explain. In this way we

find out which definitions do work for us and it is quite possible that, as we learn, several related concepts emerge with subtle technical distinctions where once we had just one. In biology, ‘heredity’ was made more precise with the term ‘gene’, which became even more precise when its relation to the nucleic acids was discovered, subsequently giving rise to even subtler concepts like ‘cistron’ and ‘operon’. The study of mechanics led to several related but distinct and well-defined concepts like velocity, speed, momentum and kinetic energy, when natural philosophy essentially had one big, fuzzy concept of ‘motion’.

The point I made above is related to the confusion between precision and accuracy. Precision is about how much margin we leave in our description, the more fuzzy, the less precise. One is accurate when one’s predictions, or otherwise claims are within the margins set by our precision. Science aims for, and progresses towards being accurate with increasing precision. When we have no way to be accurate yet, because our science is nascent at best, we should not formulate our concepts with – effectively arbitrary – precision. If we cannot see the target because we are in the dark, our estimate of its location should not be precise. However, when we send our arrows probingly, we should do so with precision so we can ascertain what aim was most accurate.

### **3.3 Nuance and sophistication**

When one presents a piece of social theory at a conference, a typical question runs something like: ‘Very interesting, but how does it address *X*? How can your theory leave out *X* while it is so obvious that *X* is highly relevant in transitions?’ For *X* can be substituted the asker’s favourite concept, e.g. ‘power’, ‘love’ or ‘networks’. Clearly, the pressure is on for the theorist or analyst to weave in *X* and ‘address’ it. This pressure is unfortunate because the question suggests a manifestly unscientific approach. The question should be whether the theory *works*, whether it explains things. If it does, without *X*, then this is an indication that *X* may not have been that important after all.

Even without such pressure there seems to be an innate tendency towards more nuance and sophistication in social theorising and this is a problem. One could object that transitions are multi-faceted processes in which not only *X*, but also the rest of the alphabet is important. That may well be, but that does not mean that a theory needs to ‘address’ all of them in one fell swoop. By conceptually overloading theory in advance, it simply becomes more difficult to figure out what really matters in what situation. It is referred to as ‘nuance’ or ‘sophistication’ but it really is vagueness, it *decreases* resolution. More of the phenomenon will fit the theory in the same way that a vague picture of a face will match more people<sup>9</sup>. There is less information – literally, see Shannon (1948) – in a nuanced picture of a phenomenon.

Simplification, by contrast, has become somewhat of a dirty word. But simplification is good. In fact, oversimplification is better. If complexity is a barrier to understanding what is going on, then simplification is part of the solution. I certainly do not mean that all aspects of transitions should be reduced to a

simple equation with concept *X*. If the mechanics of transitions truly are multifaceted – I think they are, but it is an open research question – then we should end up with many bits of theory providing piecemeal explanations based on few concepts. The grand task of future transition theoreticians is then to unify these and explain their interrelations.

One particularly problematic aspect of ‘nuance’ and ‘sophistication’ is its progressive nature, especially when combined with frameworkism (see Section 3.5). When a theoretical framework does not quite fit, the tendency is not to question the validity of the assumptions of the frame, rather, the framework is ‘improved’ by adding to it. I think this dynamic is extremely prevalent in transitions-theoretical work, see for example the response to the supposed lack of a range of letters of the alphabet in the Multi-Level Perspective in Geels (2011, p. 29–31). I am not original in my observation of the scourge of ‘nuance’ and how it thwarts progress in social theorising, see Healy’s (2017) brilliant exposé of the problem in sociological theory broadly, upon which much of the above few paragraphs is based.

### 3.4 *Not-even just-so stories*

Many of what are supposed to be explanations in transitions research really are not. Transitional phenomena are routinely ‘explained’ by means of an account of the circumstances, like a sequence of events, that lead up to them or of which they are part. While this may sound reasonable, these are often not explanations. Explanation (see Section 1.3) requires a deductive step from a general principle. If the right conditions obtain, the general principle (covering law, mechanism) holds and the phenomenon is logically entailed. Even if it concerns a proper historical explanation, where every next event is elliptically explained (i.e. by implicitly assuming a general principle) by the preceding ones, one does not necessarily obtain an explanation of the phenomenon at large.

This is perhaps best seen as a level issue. The event sequence resides on one level of explanation, while the phenomenon it describes lives on another. The set of explanations (elliptic or otherwise) of how or why one event followed another may constitute an explanation of the sequence but not necessarily of the phenomenon. The song is not explained by the string of notes. Ironically, this suffers from precisely the kind of reductionism detractors accuse scientific approaches of. Yet a narrative presenting a historical process as the logical unfolding of a sequence of events can be extremely compelling, leaving us feeling we truly understand what happened.

The reason we fall for these historicizing pseudo-explanations is psychological and the effect has been known for ages. A recent warning against it was issued by Taleb (2007) in his book *Black Swan*. He points out that we humans have a limited ability to hear a story with a sequence of events and not conclude the whole process to be explained by this. He calls it the *narrative fallacy*. It can be a proper logical fallacy, a kind of *post hoc / cum hoc ergo propter hoc* or *non causa pro causa* fallacy or the abovementioned level confusion.

A possibly nefarious consequence of this fallacy is that analysts are tempted to include only those circumstances they think ‘explain’ the outcome, that is, those that help build up a consistent, plausible narrative. Not only may they thereby be thoroughly locking in their – possibly unconscious – biases, they may prevent others to inductively form rival hypotheses since they are presented with a subset of the circumstances. In this sense, the problem is not limited to narrative explanation, though it seems extra susceptible. One straightforward way to avoid it is to present relevant hypotheses explicitly, in a testable (predictive) form, so others can at least search themselves for evidence in favour and against.

Dewey identified a related fallacy as far back as 1896 in the context of psychological research<sup>10</sup>. He called it the psychological or historical fallacy. It arises when one thinks something to be a cause or crucial factor of some state of affairs that is really an outcome of that state of affairs. For example, compare (1 – fallacy) *these things happened and this is why*, with (2 – fact) *because these things happened, this is the case*. The constitution guarantees regular elections, that is why we live in a democracy. The historical fallacy is certainly not unique to narratives, but I believe they are especially prone.

In transitions theorising it may take the following form, consider a statement like: niche innovations have difficulty to break through to the mainstream, unless the regime is destabilised by internal or external stressors<sup>11</sup>. A narrative of the events ‘explained’ by this would, for example, recount the struggle and demise of one or several innovations in the presence of a stable regime. The point is that the stability of the regime follows from the absence of niche innovations breaking through, not the other way around.

The reason I refer to all this under the header of ‘Not-Even Just-So Stories’ is that just-so stories, when used appropriately (like in adaptationist evolutionary explanations) at least explicitly present the hypothesis that is purported to explain the phenomenon in question (e.g. some adaptive trait).

### **3.5 Frameworkism**

It is rare for disciplinary scientists to not work within some theoretical frame or other and this is not necessarily problematic. Theoretical frameworks provide (1) a system for the *organisation* of data under investigation (the framework part) and (2) a *template* for explanations of phenomena to which those data pertain (the theoretical part). The merit of the first is not having to take into account everything in the world. The merit of the second is having building blocks of explanation. In physics, statistical mechanics is a framework for studying the macroscopic behaviour of gasses (amongst other things) with devices like partition functions as explanatory building blocks. In biology, evolutionary theory is framework for studying the dynamics of adaptive traits, with mechanisms of heredity and selection as explanatory building blocks. Sometimes multiple frameworks can be applied, not necessarily implying rival explanations, for example classical thermodynamics is another framework to study gasses.

There are potential downsides of any theoretical framework also. Frameworkism, where the framework part dominates the empirical research, may lead to the exclusion of data or whole cases that would provide interesting insight, for instance proving a hypothesis wrong. Ironically, such a selection bias especially shields off from empirical testing precisely the hypotheses of the theoretical framework itself. If the hypotheses underlying the theoretical frame are not questioned, for example because they are considered obvious, there is a real danger that the framework grows effectively immune to empirical testing. As a body of ‘evidence’ supporting it is amassed, there is less tendency to question it.

The dominant theoretical framework in transitions research is the Multi-Level Perspective (MLP), in particular in the form and way of application popularised by Geels since the early Noughties. Part of the allure of the MLP, according to Smith et al. (2010), is that ‘[i]ts terminology of niche, regime and landscape provides a language for organising a diverse array of considerations into narrative accounts of transitions’. More than just the language, the MLP provides a basic storyline, or in Geels’ (2011) own words, it ‘provides such a plot for the study of transitions’. The foundational assumptions of the MLP have been imported from other fields like evolutionary economics and remain basically unquestioned. For example, the assumption that the technologies and practices replacing or changing the regime in a transition always grow from niches is effectively a dogma<sup>12</sup>, whereas it should be put forward as an explicit conjecture, as a hypothesis to be tested.

### 3.6 Narrative case studies

In fact, there are two problems with narrative case studies: (1) they are narrative, and (2) they are case studies. Dawkins (1976, p. 6) concisely captures a key issue by saying: ‘I am not trying to make a point by telling stories. Chosen examples are never serious evidence for any worthwhile generalization’. But there are other issues also and, this all notwithstanding, there is surely a role for narrative case studies in transition science. I will discuss considerations about the narrative and the case study aspects sequentially.

#### *Narratives*

I already discussed some of the issues with narrative research, like an increased proneness to certain fallacies. The root of the issues with narrative research is simply its reliance on ordinary language. Avoiding the ambiguities and imprecision that come with ordinary language requires extraordinary efforts. Communication in ordinary language depends crucially on context and the other party’s ability to infer what the speaker means, usually from her own experience as a language user. This also means that the information in verbal communication is compressed (in the computer science sense of the term) relying on the possibly, likely, imperfect decompression facilities of the other end of the line.

Anyone who has ever had to make something truly unambiguously clear, for example as part of a rigorous philosophical argument, or better still a mathematical proof, knows how incredibly hard this can be. For this reason, philosophy and mathematics take recourse to formalism and technical, precisely-defined terms. The sciences also do this and it helps, see also my discussion about ‘fuzzy concepts that move’ in Section 3.2. The more precise a science needs to be in its statements, and as a rule this necessity grows as a field matures, the greater the reliance on arcane terminology and formalism and this is a good thing. Precision is important so we can be proved wrong. Ordinary language often is an enemy of precision.

A much deeper problem of the use of ordinary language became clear in the early 20th Century when philosophers realised that the structure of reality is not necessarily reflected in the structures of language. This linguistic turn was probably initiated by Russell, who used predicate logic to analyse ‘problematic’ sentences, in his ‘On Denoting’ (1905) which still stands as a point of reference in this area of philosophy. The application of predicate logic in Russell’s theory of denoting evaporated the perceived problems, inspiring Wittgenstein, Carnap and many others to investigate logic and the logical structure of the world. In other writings, Russell demonstrates how whole philosophical schools (German idealism, existentialism) are effectively based on artifacts of natural language. I am not sure whether this deeper problem has any practical consequences for transitions theorising.

I suppose we first have to deal with the ‘shallow’ problems of using ordinary language before we can even appreciate the deeper one. At any rate, we have every reason to be suspicious of narrative. Nevertheless, there seems to be a prevailing sense amongst many transitions researchers that narrative approaches are not only all right, but in fact superior. Clearly, someone has to be wrong here.

In favour of narrative, I hear such arguments as that they are better suited to deal with complexity. How this should be the case I can for the life of me not understand. If one wants to understand something complex, one should pick it apart to figure out how it all fits and works together. In other words, one should *analyse*. The more complex, the greater the need for precision and piecemeal analysis and narratives are eminently unsuited to meet these needs, as per all of the previous. If the whole could be understood as such, without breaking it down, the connotations of ‘complexity’ would not apply. Perhaps narratives are better at conveying *that* something is complex, but what of it? I do not think that is true either.

A prime example of a pro-narrative argument of this genre can be found in Andersson et al. (2014), eagerly cited by McDowall and Geels (2017). They assess the prospects of applying formal science methods to societal systems and conclude there is not much hope: ‘formal science in general would be incapable of dealing with these systems’ (p. 148). Narratives are the way to go because ‘the strength of narrative is [...] that it is the only mode of theorizing that is not obviously married to assumptions of a lack of either complicatedness or complexity<sup>13</sup> (p. 154).



This is a combination of logical fallacy and naïveté. The fallacy is this: Just because formal theorising employs strident abstractions and simplifications, does *not* mean it is married to any assumptions. Abstractions and simplifications are based on assumptions, it is true, but if the obtained formalised hypotheses explain the phenomena we have learnt something – perhaps that the simplification was warranted, and not everything mattered after all. This is not marriage, this is dating to find the true one. The naïveté is of course that narrative is much more deeply marred by assumptions, that moreover are not tested or even made explicit like they are in formalisations.

Another common genre of arguments in favour of narrative uses some sort of appeal to metaphysics. More specifically, this goes through a supposed distinction between ‘process’ and ‘variance’ theories. Variance theories explain by looking at the changes in key variables that describe the phenomenon of interest, whereas process theories explain by looking at sequences of events. How events in such sequences are described, if not as changes in key variables is unclear. Also, descriptions of a phenomenon over time in terms of a sequence of changes (events) or sequence of states (key variables) are equivalent and can always be converted into each other in principle<sup>14</sup>.

Apart from not thinking much of this distinction, I also cannot find any argument as to why one would want to use one or the other – it seems just another us-and-them distinction (we are not unscientific, we are just different). Nevertheless, the literature seems decided that process theories are the best match for transitions (e.g. Geels, 2011; McDowall and Geels, 2017). As said, sometimes the motivation is metaphysical, referring to ‘processbased ontology, which sees things as temporary instances of processes-in-motion<sup>15</sup>’ (McDowall and Geels, 2017, p. 5).

Process metaphysics goes back as far as Heraclitus obviously, but the modern exponents apparently base themselves on Whitehead, whose philosophy of process seems to draw on that of Bergson<sup>16</sup>. The Bergsonian view of change was summarised by Russell (1946, p. 763) as ‘[t]rue change can only be explained by true duration; it involves an interpenetration of past and present, not a mathematical succession of static states. This is what is called a “dynamic” instead of a “static” view of the world’. Or ‘process’ instead of ‘variance’ apparently. Bergson’s philosophy of change and time is demonstrated by Russell to be based on an error in relation to the separation between past and present, but this does not mean process metaphysics is impossible in principle.

I suppose there is not much virtue in arguing against someone’s metaphysics. It is per definition out of reach of empirical testing – otherwise it would be plain physics. And just so long as it is consistent with fact and not selfcontradictory there is no way to settle such an argument. What I *can* argue regarding process theories and metaphysics is that no-one seems able to truly analyse phenomena purely in terms of change, that is, without implicitly employing a state-based description.

Narrative, in other words, ordinary language, is supposed to be the mode for process-theoretical analyses *par excellence* but this is naïve. Narrative is just

as ill-equipped for process theory as mathematics. Always, when one verbally describes an event, some change, one describes it in terms of agents of change and objects of change: ‘Newton changed the way we think’. Newton is the agent, the way we think the object of change. Further analysis would describe how the way we think changed by describing how various aspects of it changed. Thus we are back at what essentially is a variance-theoretical description.

No matter how one tries, one describes change in terms of changing states. Ordinary language is simply unable to do it otherwise, see also my earlier remarks about the ‘deeper’ problem of ordinary language use. *All* languages have nouns and verbs and verbs take subjects (the agents of change) and sometimes objects (the objects of change). When there is no object in a declarative sentence (‘I walk’, ‘it rains’), the change would still be analysed in terms of states (‘I was there, now I am here’, ‘it was dry, now it is wet’). Perhaps a special language can be constructed in which change can be described in terms of change. My guess is that it is fundamentally impossible – perhaps someone in formal linguistics will be able to prove it as a theorem.

## Case studies

That concludes my thoughts on narratives for now. What about case studies? Case studies and narratives come as a package deal in the transitions field, it seems. Indeed, the ‘gold standard’ in empirical transitions research remains the Geelsian,  $N=1$ , narrative case study. That there are intrinsic problems with relying so much on this approach does not seem widely acknowledged. In fact, it seems to be the accepted way to go as the field’s official research agenda points out: ‘The overall direction for future transitions research involves a continued dedication to in-depth single-case research designs, also as new topics and transitions contexts are being explored’ (Köhler et al., 2019, p. 19).

Whence this dedication? And is it appropriate? What are case studies good for and good at? Although I suspect that the dedication to narrative case studies does not stem from any supposed methodological merit – it may just be another manifestation of the fetish of the peculiar – I do want to highlight some of the merits of case studies.

I think case studies (narrative or otherwise) are appropriate, if not preferred, in the roles described below. There are certainly other legitimate roles for case studies, for example with an aim of sense-making, inspiration, advocacy or providing precedent. Though these are valuable, I do not consider them to be scientific ends and therefore I shall not discuss them further. In the following, case or case study may refer to historical, narrative or any other relevant variety.

*Illustration* – This refers to situations where a case is used to clarify the meaning and application of concepts or theory. In these situations, the case is not meant as empirical evidence but as a guide to application or as demonstration of the kind of results that can be obtained. For

illustrative purposes, it does not matter in principle whether the case is factual, dramatised or hypothetical.

*Paradox resolution* – This may be a special case of illustration, where a puzzling phenomenon is shown to be a particular instance of a general principle. This is very common practice in science education. If not for illustrative purposes, paradox resolution is usually a form of inquest (see below), where a case presents a counter-intuitive phenomenon and the analyst uses *known* principles, i.e. a solid theory to explain the phenomenon.

*Inquest* – Here the aim is to learn about the specific case, to identify how circumstances or a certain course of events led to some outcome. Public inquiries, royal commissions and trials effectively conduct case studies in this vein with an emphasis on fact finding. Investigative journalism also operates in this manner. It is important to realise that the facts thus found are case specific and do not warrant generalisation on their own.

*Speculation* – Cases can suggest generalisations and in this fashion be a source for inductive hypothesis building. Especially if the case is typical for some class of phenomena or if there are numerous cases in which the same patterns are observed, this can make a strong case for a hypothesis. The caveat is of course the status of the hypotheses thus obtained. They are *conjectures*, and should be explicitly designated as such. One cannot test hypotheses with the same data that suggested them, fitting the existing data is but a *minimum* criterion for a hypothesis.

The only other role I see for the kind of narrative case studies so prevalent in transitions research is as part of a systematic effort towards the building of a large empirical dataset covering very many cases. As long as the concepts are still fuzzy and the mechanisms are unclear, cataloguing cases in great detail is not only appropriate but also desirable, but it comes at a cost. Systematically using large amounts of qualitative, especially narrative, data is hard. Just as it is hard to systematically catalogue such data in order to make them available programmatically, i.e. using software. But just because it is hard should not mean we should settle for an unscientific approach. I will discuss this further in the concluding section. See also my Chapter 12 with Martínez Arranz and Spekkink in this volume.

#### 4 Theory, models and empiricism

In this section, I will argue that central to scientific practice is an interplay between theory, models and empirical data. Science, roughly speaking, aims at understanding by means of constructing *theories* of classes of phenomena. Science also, again roughly speaking, is *empirical*, meaning that observation and experiment provide the inspiration and touchstone for scientific knowledge. Experiment is the conscience of science, theory her soul. Models, I claim, perform a *mediating* role between theory, empirical data and human understanding. And they do so in several different ways.

I should clarify upfront that my notion of ‘model’ is much broader than the one the rest of this book employs, which refers mainly to simulation models. The sort of model that I am particularly interested in here is what could be called ‘scientific models’, by which I mean models that are instruments in the context of scientific investigation. This notion ranges from Bohr’s model of the atom, through to Iterated Prisoner Dilemmas as models for the evolution of altruism and Lotka–Volterra style equations for population dynamics, all the way to computer models like in agent-based social simulation.

Given that I claimed models to mediate between three things, there are in fact three distinct mediating roles for models. I will treat them sequentially, but before I do so, I will need to digress briefly on the character of theory.

#### **4.1 Theory, models and meta-models**

I am assuming, perhaps foolishly, that the notions of ‘empirical data’ and ‘understanding’ are sufficiently clear and that the examples I gave of the kind of models I mean will suffice for now. To assume such things about ‘theory’ is dangerous, so an explication for the purposes of this chapter is appropriate.

My views on what theory is and does are an amalgam of several that are all in the mainstream of philosophy of science. For the interested, these are views ranging from classics, like Carnap (1956) and Hempel (1966), to more contemporary central figures, like Salmon (1990) and van Fraassen (1980) and several others. The issues on which these and other authors disagree, or where I disagree with them, are either not relevant here or about technical philosophical matters, e.g. whether one should be realist or anti-realist about theoretical concepts. It is in this sense I stated in the introduction that my views are philosophically uncontroversial.

Theories are both the repositories of knowledge acquired and the key to knowledge new. As a repository of knowledge, a theory contains a collection of what could be called ‘templates of explanation’ about a certain class of phenomena. These templates, depending on the theory, may be formal or fuzzy but at any rate they are generic. Applying theory, then, consists of taking a template and substituting the particulars of the case at hand for the generic terms in the template. This application is precisely what modelling is.

For example, Maxwell’s theory of electrodynamics explains the class of phenomena related to electricity, magnetism and optics. The theory provides an explanatory template in the form of four partial-differential equations (the Maxwell Equations) and one additional equation (the Lorentz Force Law). For any particular case (hypothetical or actual), one provides the charge distributions, which determine the fields (or the other way around) and then explains the phenomenon by mathematically deriving it as a prediction. Describing the case context (charge distributions) with the explanatory template (Maxwell’s equations) of the theory constitutes a model.

In the above case, the model is fully particular but this need not be the case at all. Clearly, explanatory templates can be set-up for commonly recurring empirical situations by equipping them with only those particulars these

situations have in common. Or similarly, for situations that are variations on a theme, the particulars can be parameterised. The models thus obtained are of course closer to the common understanding (if there is such a thing) of the term ‘model’, though the term ‘meta-model’ would perhaps be more apt. Such meta-models apply to a class of phenomena that is a subclass of the phenomena the theory is about. This does not exclude the possibility that the (meta-) model applies to various phenomena outside of the scope of the theory also. This is so exceedingly common that it is in itself a motivation for model-based research.

Taking the above view to its logical conclusion, one obtains that theories themselves are meta-models, or patchworks of meta-models. Here also, one sees the aspirational nature of formal theory. While our understanding is dim our theory may be formulated in fuzzy terms, allowing various operational definitions and corresponding models. As our understanding gains accuracy, the precision of our concepts can gain concurrently.

#### 4.2 *Between theory and understanding*

This is perhaps the most abstract of the mediating roles. I suspect many theoreticians in the sciences employ it intuitively without questioning it or even be aware of it being at work. Working with theory always entails working with more or less specific models based on that theory. This holds whether the theory is established or in development.

Using an established theory to inform one’s general understanding of phenomena is based on creating and manipulating *ad hoc* (meta-) models of various situations the theory is supposed to cover. In a very literal sense, this is how students in the mathematical sciences learn theories, working through scores of problems with pencil and paper. In a less explicit, but still literal, sense this is how everyone learns theory, also when it is not-so-formal. Having ingested the raw material of a theory, one mentally tries it on familiar cases and tests it on apparent exceptions. Developing theory makes use of the very same creating and manipulating of *ad hoc* (meta-) models.

The more explicit (tentative) models are formulated, the more they lend themselves for active exploration. Exploration can mean thought experiments, where one tries to take one’s assumptions to their logical extremes, either to find the limits of applicability or simply see what would happen. Using simulation, a model can be ‘run’ to produce patterns over time, showing how one’s nascent theory behaves in ways that the naked mind is unable to. Another virtue of explicit models, especially when formalised is that they enable checking consistency, both internally and with existing, established theory.

It should be noted that models can, often do and perhaps even should, involve deliberate simplifications and distortions of what one knows to be the case in ‘reality’. Common examples are ‘course graining’, where one ignores dynamics on smaller scales (e.g. by looking at whole-of-industry dynamics) or idealisations, where certain phenomena are ignored (e.g. frictionless pendula). Even when such ignoring seems utterly unjustified (friction often does matter) such

models are still useful. In fact, they can be better than models taking many such factors into account because they look at the *pure* phenomenon, hence they are called idealisations. Holland (2006) emphasises the use of toy models, which are deliberately oversimplified models, to learn about complex, adaptive systems.

### **4.3 Between empirical data and understanding**

Making sense of empirical data entails viewing them as particular instances of generic categories. This holds both in quantitative and qualitative contexts. An example of the former is counting adoptions of a certain technology. An example of the latter is identifying something as a type of event like a government intervention, or a type of actor like a frontrunner. Understanding comes from finding patterns among the data and then capture those patterns in terms of hypotheses. Again, both when capturing data in generic categories and in the capturing the patterns in hypotheses, one is making models.

Close-to-data models like these may be truly *ad hoc*, serving only to see whether there is any pattern at all, or to figure out whether a certain theory or other model can fruitfully be applied. Such models may be evanescent mental constructs, never even written up, while ‘eye-balling’ an interview transcript, say, or they may be based on formal ontologies<sup>17</sup> and employ statistical techniques.

It is not always necessary to provide the categories or concepts *a priori*. Standard methods in statistics like factor analysis or principal component analysis enable the identification of relevant categories on the basis of the data themselves, this is commonly used in psychological research.

### **4.4 Between theory and empirical data**

This is perhaps the most straightforward role of models and in a sense the short-circuiting of the previous two roles. This role works in two directions, namely: (1) models enable empirical data to be explained by framing them in the terms of the theory, and (2) they enable the derivation of predictions from the theory which can be tested against empirical data.

## **5 Conclusion – concrete steps towards transitions science**

I would like to conclude this chapter by suggesting several, relatively concrete, things that could be done to move towards a science of transitions. Some of these are being done already but none seem to be anywhere near the norm in the field. For the sake of overview, I will group my suggestions under a theory and data related header, despite their interrelatedness.

### **5.1 Theory development as modelling and vice versa**

Taking my view of theory as meta-model at face value (see Section 4.1), I think theory development can progress more, and more systematically if it is

approached as modelling. I do not mean that everyone should write computer models, nor that such modelling is all we should do. I mean an approach to theorising where hypotheses are explicitly framed, preferably in a case a-specific format, in such a way that consequences can be derived from them. At the very least, that will enable straightforward hypothesis testing against data. More ambitiously speaking, it makes them amenable to formal modelling.

Such an approach has been advocated for the social sciences by people like Hedström and Swedberg (1998, see also Hedström 2005). They refer to hypotheses framed in this fashion as ‘social mechanisms’ and see a significant role for (computer) modelling. In Chapter 5 of this volume, Bianchi & Squazzoni also discuss this view. Ideally, any candidate mechanism is deployed in several model forms, for example based on different operational definitions of the concepts involved and using various modelling approaches, ranging from verbal descriptions with rigorous definitions, through to mathematical formulae or simulation models. All of these can then be explored as toy models, as part of other models and, of course, tested against empirical material. In a slogan: *one mechanism, multiple models, many data*.

Though not a strict necessity to do so, many hypotheses actually can be framed in one or more semi-quantitative forms. I say ‘semi’ because I am not referring to metrics or indicators. What I mean is that a phrase like ‘frontrunners play an important role in early phases of transitions’ can be phrased as ‘more frontrunners advocating practice X, lead to an increased early uptake of that practice in comparison to cases where they are absent’. In a format like the latter version, correlations can explicitly be investigated. Moreover, most hypotheses will have several of these semi-quantitative versions.

I have already discussed some of the virtues of formalisation in Section 4. The trajectory of the natural sciences has invariably been towards more formalisation as they progressed. Like with the argument about precision, I cannot infer that formalisation is behind that progress. My guess is that it did help and one of the factors would have been that it adds deductive argument and mathematical proof to the toolbox of the theoretician. Formalisation should not be confused with quantification, in other words concepts do not have to correspond to quantities like in physics. Linguistics, as a science, essentially begins with Chomsky’s formalisation effort, which was a formalisation of the structure of language.

## 5.2 Dealing with data differently

Though narrative case studies have a role in scientific transitions research, different ways of engaging with empirical material will be necessary. Moreover, altogether different sources of data need to be harnessed. This, in combination with the model-based theorising I discussed above, will – so I argue – greatly strengthen the approach.

There is really no strong reason to single out the case as the preferred unit of analysis. Ideally, the kind of empirical data to use should be informed by the

hypotheses we are trying to test. In this regard, it can be instrumental to think in terms of data *sets* rather than case data. The epilogue of this volume discusses in some detail the kinds of data sets that would be relevant for scientific transitions research. To give an idea, one thinks of financial flows and how they change, text mining of news articles or twitter feeds, energy use of industries, morphological patterns in built environments, organisational networks, statistics of organisational life cycles (founding, merging, death), social networks of key transition actors and so on and so forth. There are many kinds of data from which to learn about transitions that we are not even considering.

One way in which case-oriented research may have continued relevance is through meta analysis. Fields like medicine and psychology, similar in complexity of subject, routinely use meta analysis to check findings across many cases. This helps resolve conflicting hypotheses based on different cases and find anomalies. Using the considerable volume of transitions case studies for meta analysis would allow testing key assumptions and new hypotheses. Ideally, existing case studies would be cast into a form in which they can be queried and then made available as a database for the entire research community. With Martínez Arranz and Spekkink (Chapter 12 of this volume), I discuss meta-analysis, case databases and other data-driven methods for transitions research in more detail.

Letting go of the case-orientation, dealing with data differently and more effectively is not going to be easy for two reasons. The first of which is that it simply is not easy. Transitions as a research topic are not special in this sense, constructing good data sets is difficult in any science and careers are built on devising better ways to obtain data. I am always affronted by remarks to the effect that the natural sciences have it easy because their data are quantitative. Or because their topic is ‘not complex’. It is in fact incredibly difficult to design experiments that isolate a phenomenon from the complexities it is entangled in. Saying that this is impossible for transitions is just giving up in advance.

The second reason that it is going to be difficult to deal differently with data is because few transitions researchers are currently conversant with the methods and techniques this will require. And a prevailing, but misplaced, attitude of superiority regarding the dominant research practice is not going to do the field any favours in this respect. Not everyone needs to become a modeller, but basic mathematical and programming skills may become indispensable. At any rate, reading secondary sources and synthesising them into a multi-level narrative will no longer suffice to make progress. But there is a reward, the systematic availability of new, preferably large, data sets, will allow the use of tools now mostly out of reach. Notably, a rich suite of statistical and visualisation methods will become available. Not to mention the improvements in the robustness of the knowledge produced.

In conclusion, we should acknowledge that despite much empirical work, we do not know much about transitions scientifically yet. I think there is no shame in that and I am optimistic about the prospect of a transitions science. In the meanwhile, when we find ourselves in a position to influence decisions



based on our knowledge of transitions, we should ask ourselves what gives that knowledge the authority and legitimacy to be such a basis. If it is not science, then what is it?

## 6 Acknowledgements

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## Note

- 1 Wikiquote says it is not in any of the Feynman books. I have a relatively clear but quite probably false memory of reading the quote as an undergrad.
- 2 See also my Chapter 12 with Martínez Arranz and Spekkink (this volume) on this comparison in relation to meta analysis.
- 3 See Section 3.2 about the important distinction between accuracy and precision.
- 4 People who are irked by my repeatedly citing Feynman, will see this as yet another sign of my narrow, physics-based views. Perhaps they are right. Either way it does not invalidate those views. I cite Feynman because of his talent to state in simple but not simplistic terms the common sense of science. To my delight, Dennett, on this topic of publishing 'negative' results, cites the very same Cargo-Cult lecture by Feynman.
- 5 Cargo cults are a Melanesian phenomenon that has attracted considerable anthropological attention. Melanesian people observed aeroplanes landing with cargo bringing material wealth. When the aeroplanes no longer came, they began imitating the practices associated with the arrival of cargo, for example by building ritual landing strips. Most cargo cults seem to have disappeared since the 1950s though a few persist. Wikipedia has several lemmas on cargo cults, see [https://en.wikipedia.org/wiki/Cargo\\_cult](https://en.wikipedia.org/wiki/Cargo_cult), and see Otto (2009) for an overview article of the anthropology.
- 6 Using back-of-the-envelope calculations of scale-ranges for mass, surface area or the numbers of individuals in societies, I estimate the systems of society to cover less than 10 orders of magnitude.
- 7 There is another, very specific, meaning of 'ontology' in computer science that is related to, but different from the usual philosophical meaning.
- 8 See Dennett (2013) for an interesting discussion about such commonly used but metaphysically difficult notions.
- 9 See Russell (1923) on vagueness in this sense, including an example much like the picture one here.
- 10 See also the American Psychological Association's online dictionary at <https://dictionary.apa.org/historical-fallacy>.
- 11 An example of this reasoning (not so much the fallacy) can be found in Geels (2002, p. 1272): 'Radical innovations, which are pioneered in niches, have a hard time to break out of the niche-level. If the regime is confronted with problems and tensions emerge,

- the linkages in the configuration “loosen up”. The configuration becomes “warm” (Callon, 1998). This creates opportunities for radical innovations to escape the niche-level and be incorporated in the ST-configuration’.
- 12 As can be judged from phrases like ‘niches as the locus where radical variety is generated, and regimes as selection and retention mechanism’ (Geels, 2002). The retort in Geels (2011) is that there may have been a bias towards bottom-up dynamics and that more attention needs to go to regime and landscape processes. I do not care if it is a bias, it may well be true. The point is it needs to be tested. As regards the bias, in Geels et al. (2016) all the pathways are still described in terms of niche innovations replacing or altering the regime.
  - 13 They do add that this ‘does not mean that it is particularly good at dealing with their presence’.
  - 14 See my detailed argument for this in my Chapter 12 with Martínez Arranz and Spekkink in this volume.
  - 15 In reference to a distinction between ‘strong’ and ‘weak’ process-based ontologies made by Langley (2009)
  - 16 Russell wrote: ‘His philosophy in later years was essentially that of Bergson’ (Feinberg and Kasrils, 1969, p. 160).
  - 17 These are formalisations of knowledge domains or ‘universes of discourse’, see for example Uschold and Gruninger (1996) and Gruber (1993). Though related, this usage of the term ‘ontology’ is distinct from its use in philosophy. See also Footnote 7.

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